

EE143: Microfabrication Technology

Spring 2026

Lab Section: Wednesday 9-12 PM

Group 20

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GSI: Shiwoo Lee

Professor: Ali Javey

Contributions:

Huan: Summary [I], inputted most values, calculated the Overetch and Misalignments, adding screenshots, wrote some step descriptions for Experimental Details [II], calculated oxide thicknesses (using graphical and two math approach) for Theoretical Calculations [III], filled out feature size deviation, misalignment, and oxide growth tables for Process Deviations [IV], answered Q3-6 for Questions [V].

Prathamesh: Created all of the MOSFET cross-section and top-down drawings and wrote many of the step descriptions for Experimental Details [II], calculated dopant distributions and sheet resistance for Theoretical Calculations [III], and coded a computational model to verify results [III]. Formatted the final document.

Yarizmar: Provided the data for sections I and II for Wafer #20, did the explanations for section IV, and theoretical calculations for aluminum sheet resistance in both sections [III & IV], did Questions [V] Q1-2, 6 to 10, and revisions [I, II, III, IV, V].

I. Summary

The goal of the fabrication process was to fabricate a p-type silicon wafer into an NMOS integrated circuit containing MOSFETs, logic gates, and test structures that are functional. Throughout the process, we encountered some issues, ranging from one of the [HF] solutions not working properly and not etching the wafer well, residue being left on the wafer (contamination problem), and even the wafer being burnt on the edges during the Metallization process. On the other hand, we were able to successfully align the patterns accurately during the exposing process, etching for the most part went smoothly, and photoresist spin coating processes were mainly successful, except for our second time working with the photoresist, which we had to dissolve and start over, as it did not cover the wafer adequately. After completing the fabrication process, we learned that each step significantly affected the device's performance. We have to think carefully about every step, like whether we should overetch or underetch, since the device can be sensitive to any changes. Acids have to be precise, and timing is imperative. Checking measurements after each step is important since errors could propagate in future steps, and contamination will most likely happen.

II. Experimental Details

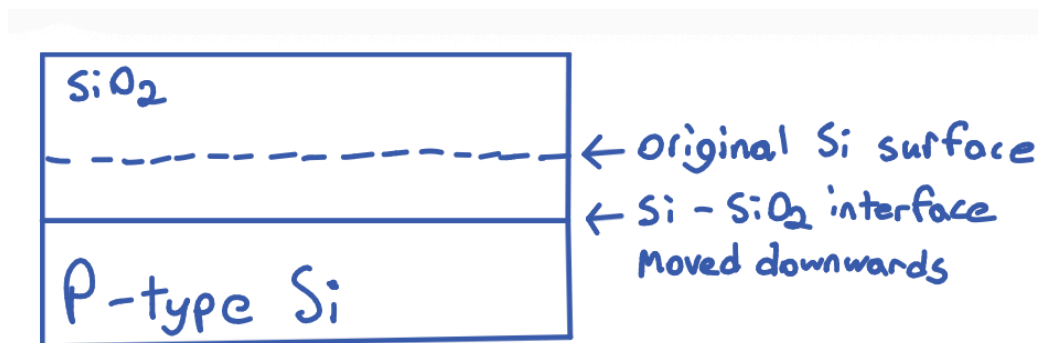
Fabrication Steps:

Initial Wafer + Field Oxidation

Describe Step: Starting with the p-type silicon substrate, our lab TAs grew a thick layer of silicon dioxide over the surface through thermal oxidation. The oxide acts as an insulator that isolates components on the wafer and defines where transistors can and cannot form to prevent parasitic effects.

Parameter	Value
Wafer orientation	$\langle 100 \rangle$ crystal orientation
Type of dopant (atom and n-type vs p-type)	B11: Boron = p-type
Field oxidation temperature ($^{\circ}\text{C}$)	1000 $^{\circ}\text{C}$
Field oxidation time (minutes)	5-125-5 minutes
Field oxidation type (wet or dry)	Dry-Wet-Dry [Main growth is in Wet]
Measured field oxide thickness (nm)	630 nm
Measured sheet resistance (Ω/square)	370 Ω/square

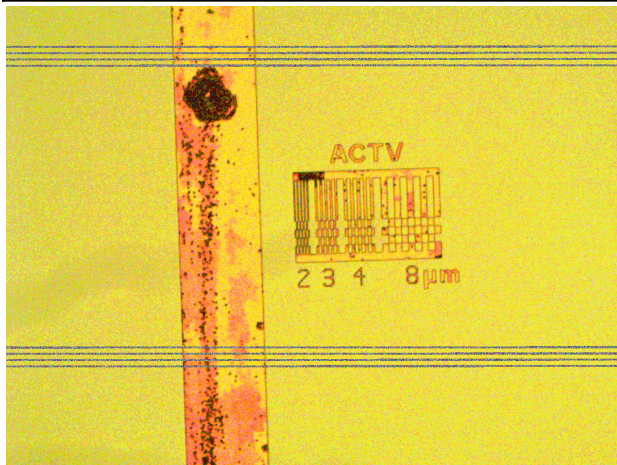
Initial Wafer + Field Oxidation MOSFET Cross-Section Drawing:



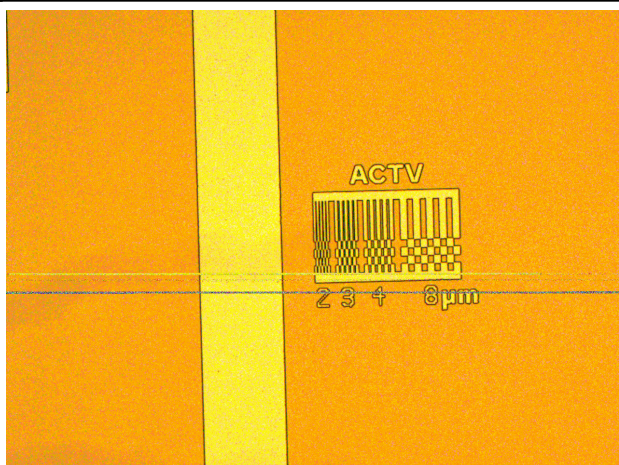
ACTV Photolithography + Etch

Describe Step: The test consists of spinning photoresist onto the wafer, exposing the wafer using the specific ACTV lithography mask, and then etching the field oxide using the photoresist as a mask. This layer is important for patterning the oxide to define where current will be allowed to flow (the places where the oxide is removed).

Parameter	Value
Dehydration bake temperature (°C)	100°C
Dehydration bake time (min)	10 min
HMDS time (min)	5 mins
Photoresist spin speed (RPM)	3000 RPM
Photoresist spin time (s)	30 s
Soft bake temperature (°C)	100°C
Soft bake time (s)	90 s
Exposure time (s)	7.5 s
Development time (s)	~ 60 s
Etch time (s)	7 mins = 420 s



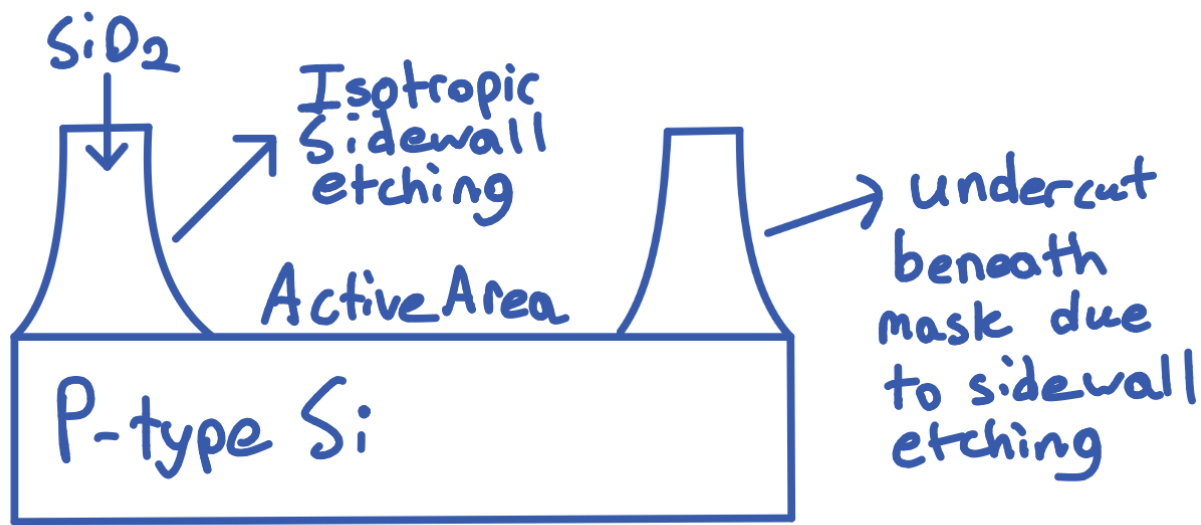
After lithography



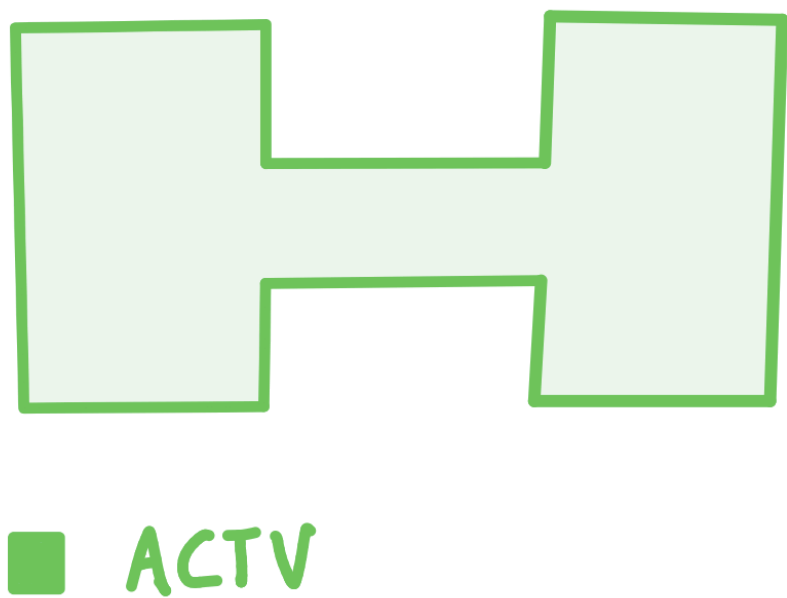
After etch + strip

Overetch (μm)	2 μm	3 μm	4 μm	8 μm	Average
After lithography	-0.0195 μm	-0.114 μm	-0.073 μm	-0.18 μm	-0.0966 μm
After etch + strip	0.086 μm	-0.353 μm	0.532 μm	0.711 μm	0.244 μm

ACTV Photolithography + Etch MOSFET Cross-Section Drawing



ACTV Photolithography + Etch MOSFET Top-Down Picture:

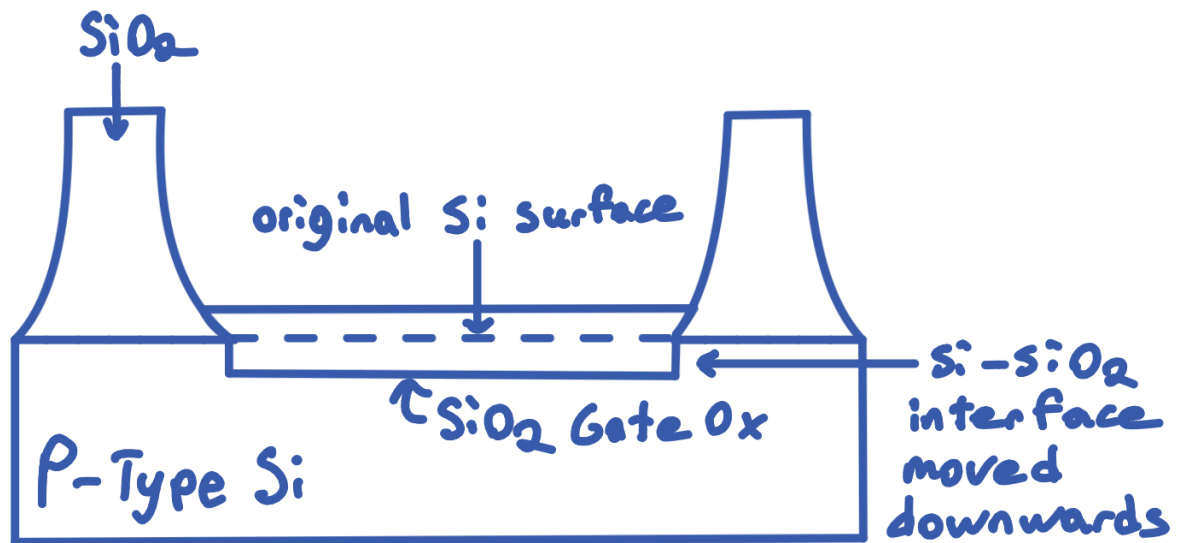


Gate Oxidation

Describe Step: This step consists of using a furnace to dry-oxidize the wafers to create the gate oxide, and also includes an annealing step to allow silicon atoms to diffuse and heal the damage done during oxidation. The gate oxide is extremely important to the operation of the transistors, as it acts as an insulator between the gate and channel. It is also crucial that the thickness of the gate oxide is controlled, as too thick an oxide will cause weaker gate control, while a thinner oxide will cause leakage.

Parameter	Value
Furnace temperature (°C)	1050°C
Furnace time (minutes)	40 mins
Oxidation type (wet or dry)	Dry
Measured gate oxide thickness (nm)	110 nm

Gate Oxidation MOSFET Cross-Section Drawing:

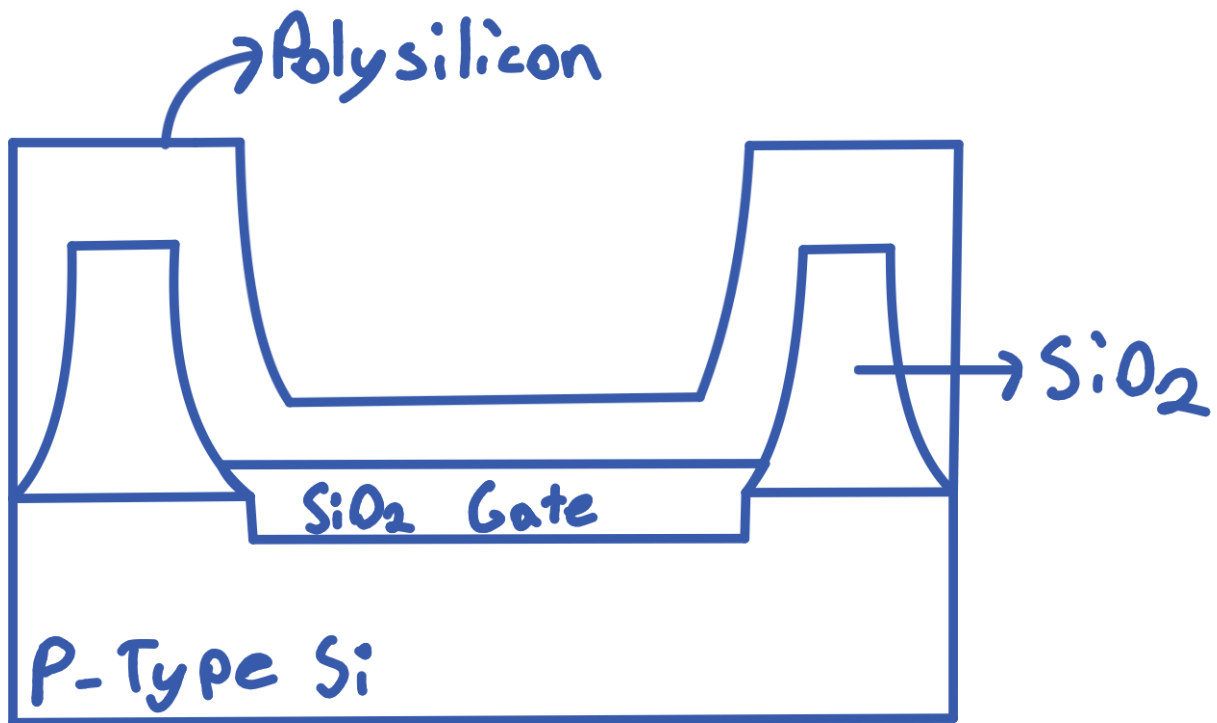


Polysilicon Deposition

Describe Step: The polysilicon deposition step involves depositing polysilicon doped with phosphorus on top of the wafer using the tystar10 machine, with silicon coming from silane and phosphorus from phosphine. This step creates the basis for the conductive gate electrode that will control the flow of the current when voltage is applied, and the layer of doped polysilicon will be patterned in later steps to form the transistor gate.

Parameter	Value
Furnace temperature (°C)	615°C
Furnace time (minutes)	3.5 hours
Measured polysilicon thickness (nm)	650 nm

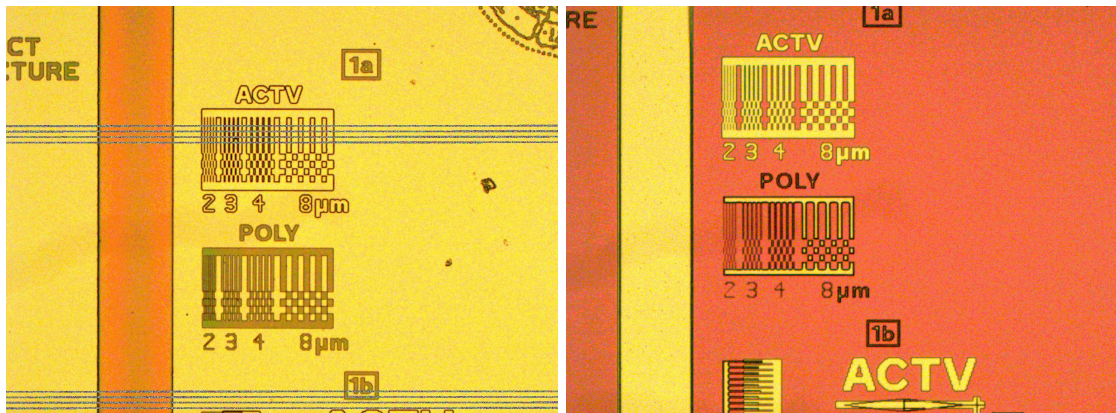
Polysilicon Deposition MOSFET Cross-Section Drawing:



POLY Photolithography + Etch

Describe Step: This step consists of applying photoresist to the wafer that is now covered in a layer of oxide and polysilicon, and then exposing it using the POLY lithography mask. Following this, we first etch the polysilicon layer with silicon etchant to pattern the polysilicon into the gates that will be probed, and then use HF to etch the areas that will become the source/drain areas. This results in a wafer with properly patterned gates that can now control the flow of current, and also contains cleared areas for the source and drain areas to form in the next step.

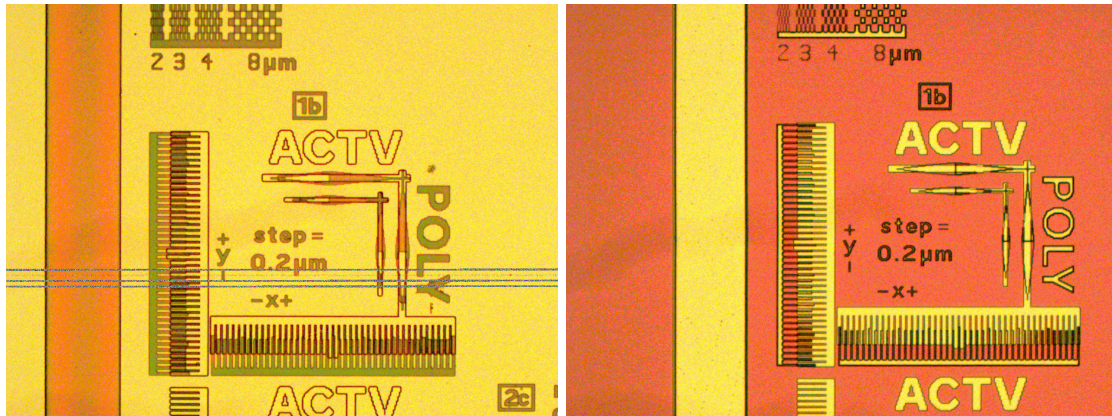
Parameter	Value
Polysilicon etch time (s)	3:30~4 min
Gate oxide etch time (s)	1 min



After lithography

After etch + strip

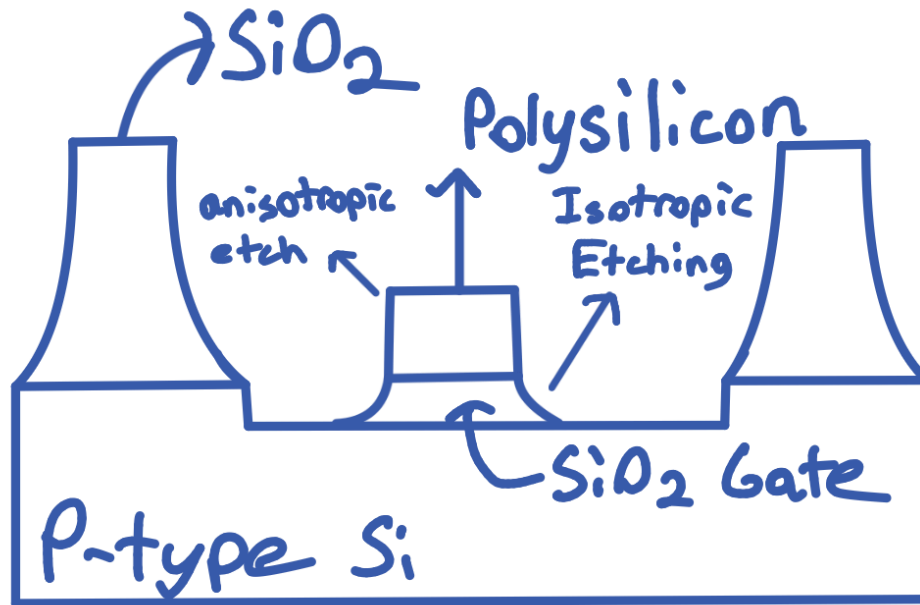
Overetch (μm)	2 μm	3 μm	4 μm	8 μm	average
After lithography	0.112 μm	0.288 μm	0.118 μm	0.237 μm	0.189 μm
After etch + strip	0.058 μm	-0.191 μm	-0.412 μm	-1.278 μm	-0.456 μm



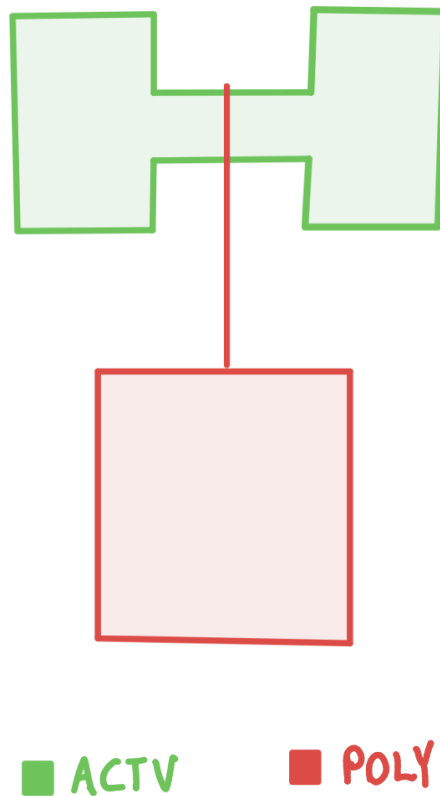
ACTV/POLY alignment mark

X misalignment (μm)	Y misalignment (μm)
0.6 μm	0.4 μm

POLY Photolithography + Etch MOSFET Cross-Section Drawing:



POLY Photolithography + Etch MOSFET Top-Down Drawing:

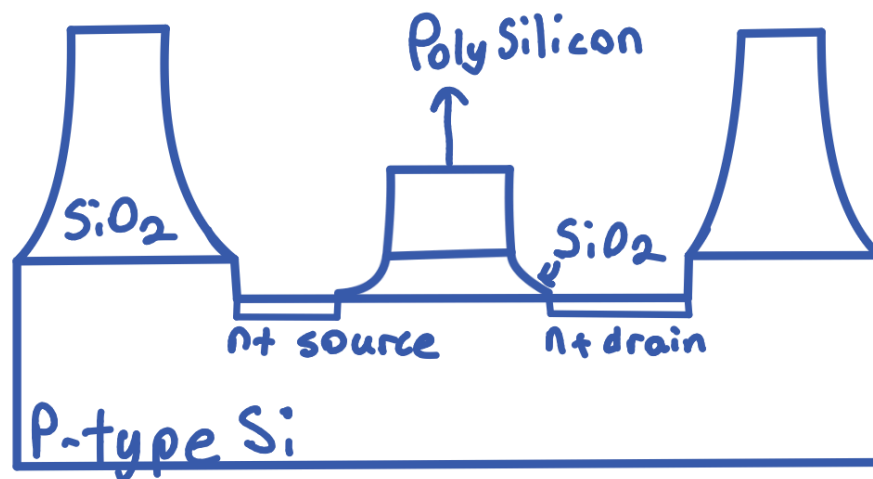


Source/Drain Doping + Intermediate Oxidation

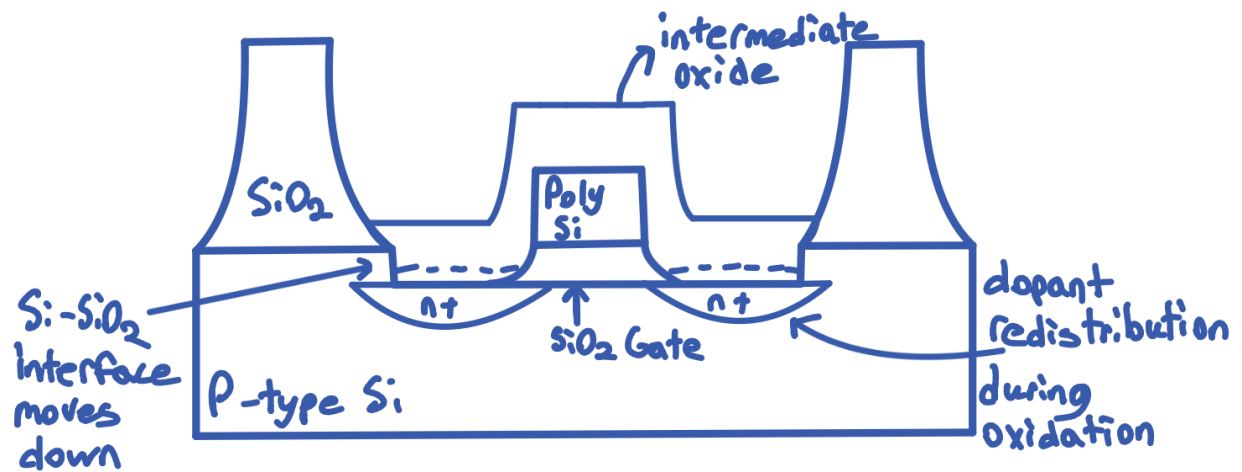
Describe Step: The source/drain doping step involves applying and baking spin-on glass (SOG) coating, which contains a high concentration of phosphorous dopants that are diffused into the wafer through the furnace. The exposed areas of the silicon are now doped n+ source and drain regions after this step, and are separated by the channel located underneath the gate. Following this, we have the intermediate oxidation step, which consists of driving the dopants deeper into the silicon to form the proper source/drain junctions on either side of the channel while growing an intermediate oxide layer through wet oxidation. Both the dopant drive-in and the intermediate oxide growth occur using the furnace (the wet oxidation is facilitated by the bubbler).

Parameter	Value
Doping furnace temperature (°C)	1050°C
Doping furnace time (min)	10 min
Type of dopant (atom and n-type vs p-type)	Filmtronics Phosphorosilica or Phosphorus (P), n-type
Drive-in furnace temperature (°C)	1050°C
Drive-in furnace time (min)	37 min
Oxidation type (wet or dry)	Wet
Measured intermediate oxide thickness (nm)	55 nm
Final control wafer sheet resistance (Ω/square)	4.82 Ω/square

Source/Drain Doping MOSFET Cross-Section Drawing:



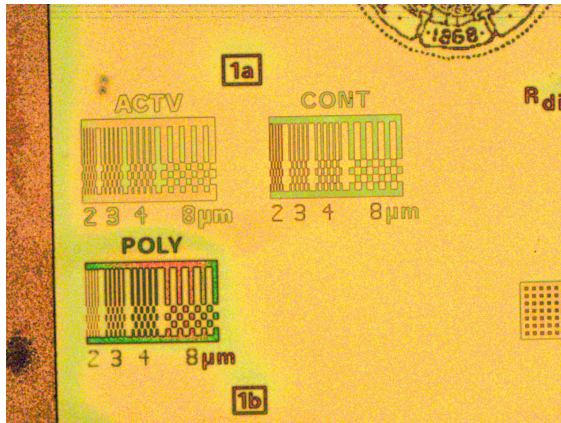
Dopant Drive-In and Intermediate Oxidation MOSFET Cross-Section Drawing:



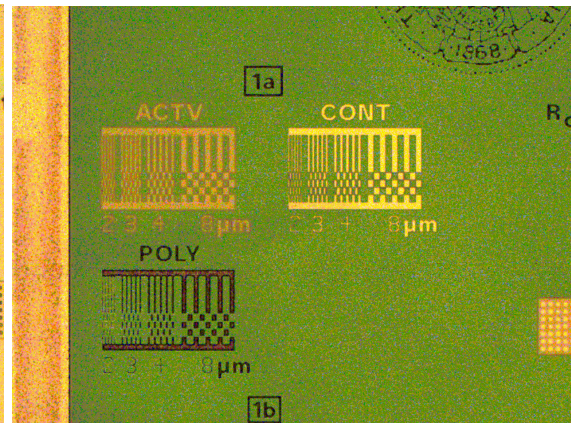
CONT Photolithography + Etch

Describe Step: This step consists of exposing the applied photoresist using the CONT mask and etching with BHF. Following the etching process, the wafer now has contact holes that provide the metal layer access to the gate, source, and drain regions.

Parameter	Value
Oxide etch time (s)	1:30 mins = ~90 s

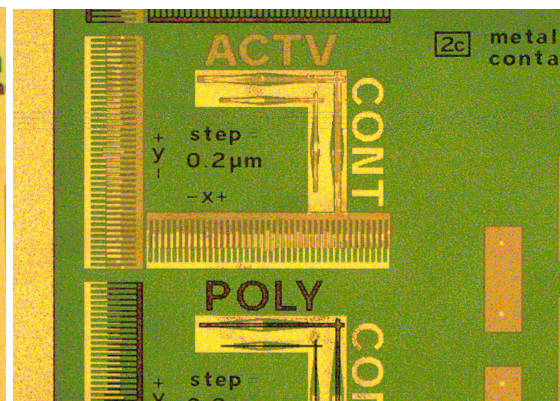
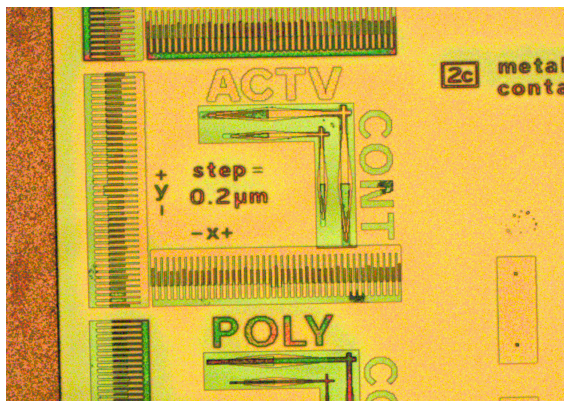


After lithography



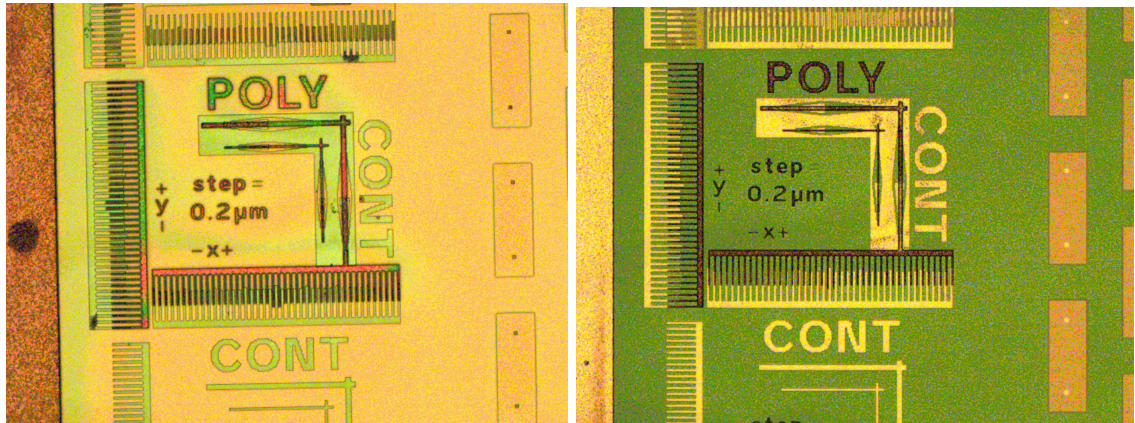
After etch + strip

Overetch (μm)	2 μm	3 μm	4 μm	8 μm	average
After lithography	-0.091 μm	-0.209 μm	-0.333 μm	-0.493 μm	-0.2815 μm
After etch + strip	-0.48 μm	-0.099 μm	-0.102 μm	-0.315 μm	-0.249 μm



ACTV/CONT alignment mark

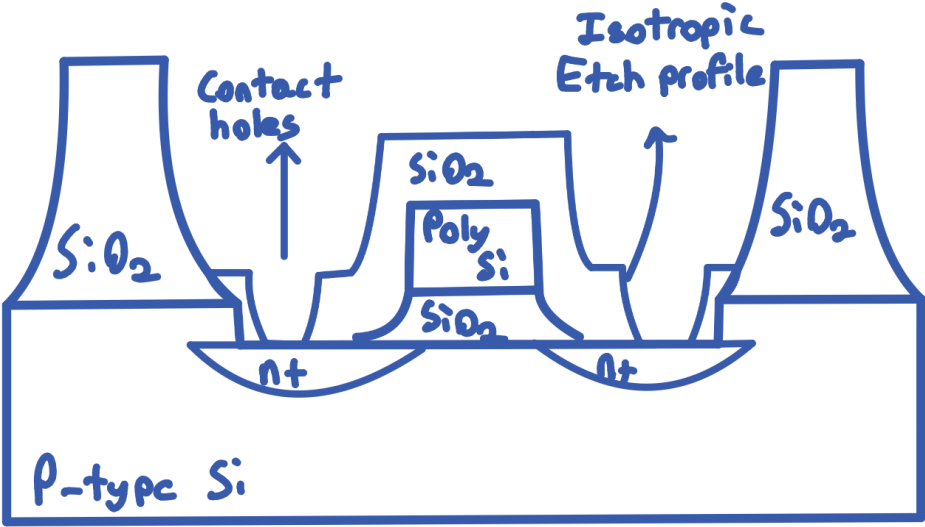
X misalignment (μm)	Y misalignment (μm)
0.2 μm	0.2 μm



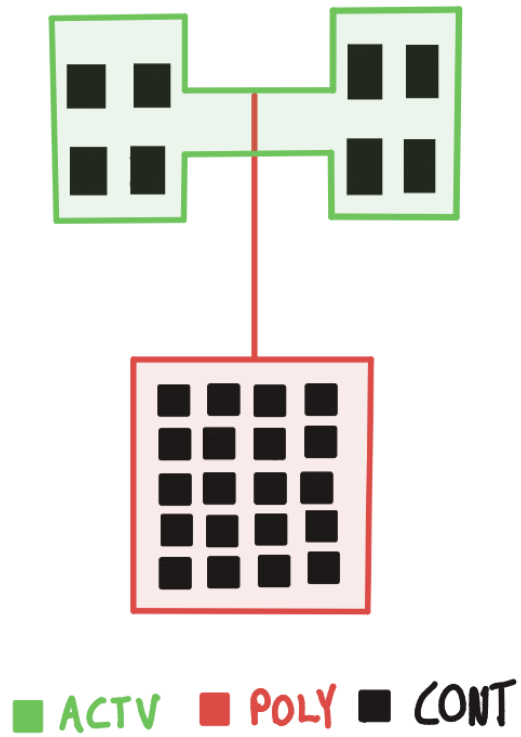
POLY/CONT alignment mark

X misalignment (μm)	Y misalignment (μm)
0.2 μm	0.2 μm

CONT Photolithography + Etch MOSFET Cross-Section Drawing:



CONT Photolithography + Etch MOSFET Top-Down Drawing:

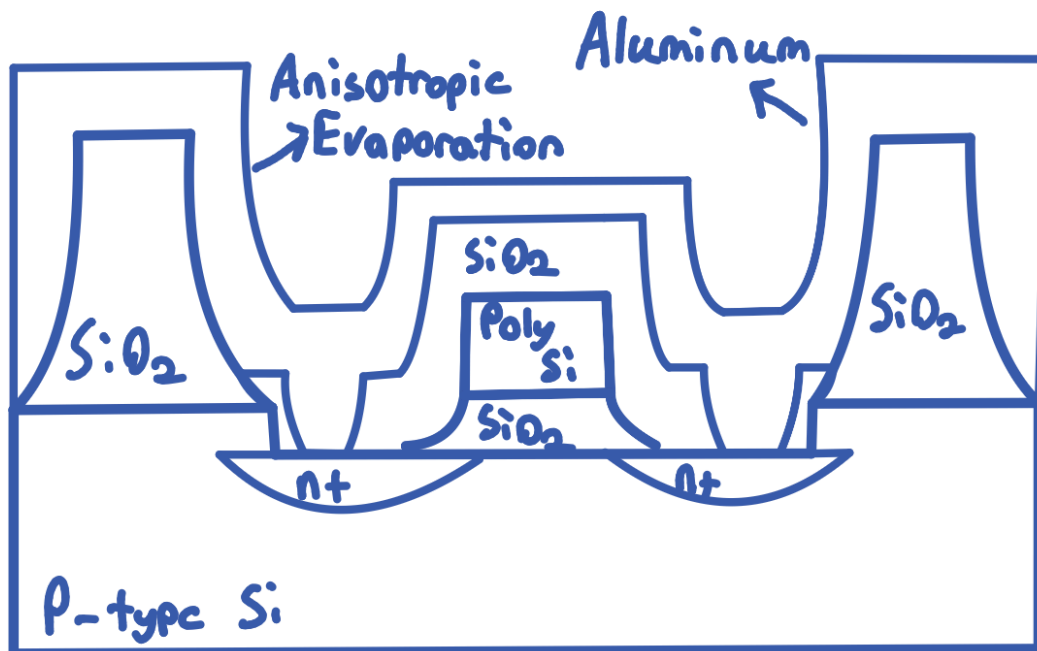


Metallization

Describe Step: This step consists of the evaporation of aluminum to create a deposited aluminum layer. The evaporation is conducted using the evaporator, and the aluminum thickness is measured through the four-point probe. This step results in a conductive aluminum layer on top of the wafer that forms electrical connections to the source, drain, and gate through the contact holes created in the CONT lithography and etch step.

Parameter	Value
Aluminum sheet resistance (Ω/square)	$1.973 \times 10^{-1} \Omega/\text{square}$

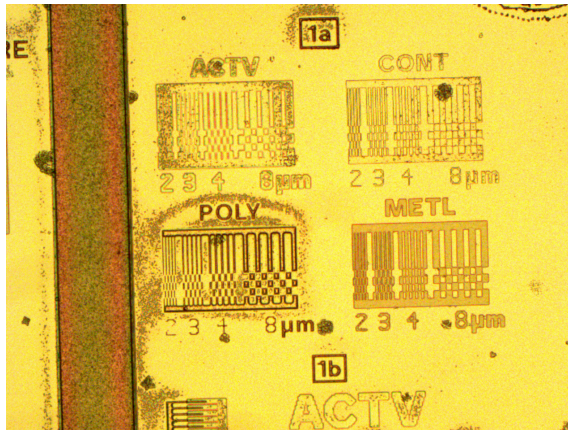
Metallization MOSFET Cross-Section Drawing:



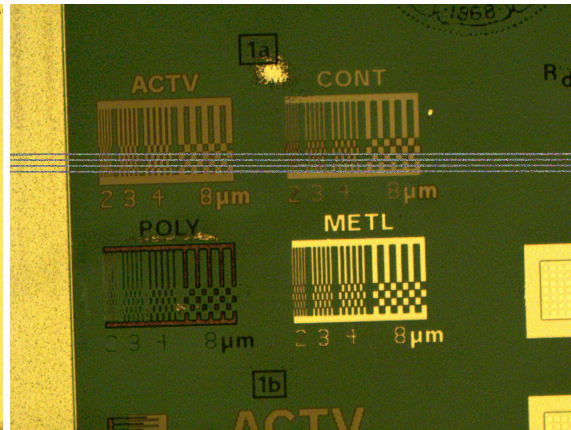
METL Photolithography + Etch

Describe Step: This step consists of exposing the applied photoresist using the METL mask and then etching the aluminum using Al etchant, followed by sintering, which allows the interdiffusion of the silicon and aluminum to form a good contact and stabilize the oxide-channel interface. Following this step, the metal interconnects to the gate, source, and drain are now defined.

Parameter	Value
Metal etch time (s)	45 seconds
Sintering furnace temperature (°C)	400°C
Sintering furnace time (min)	20 mins

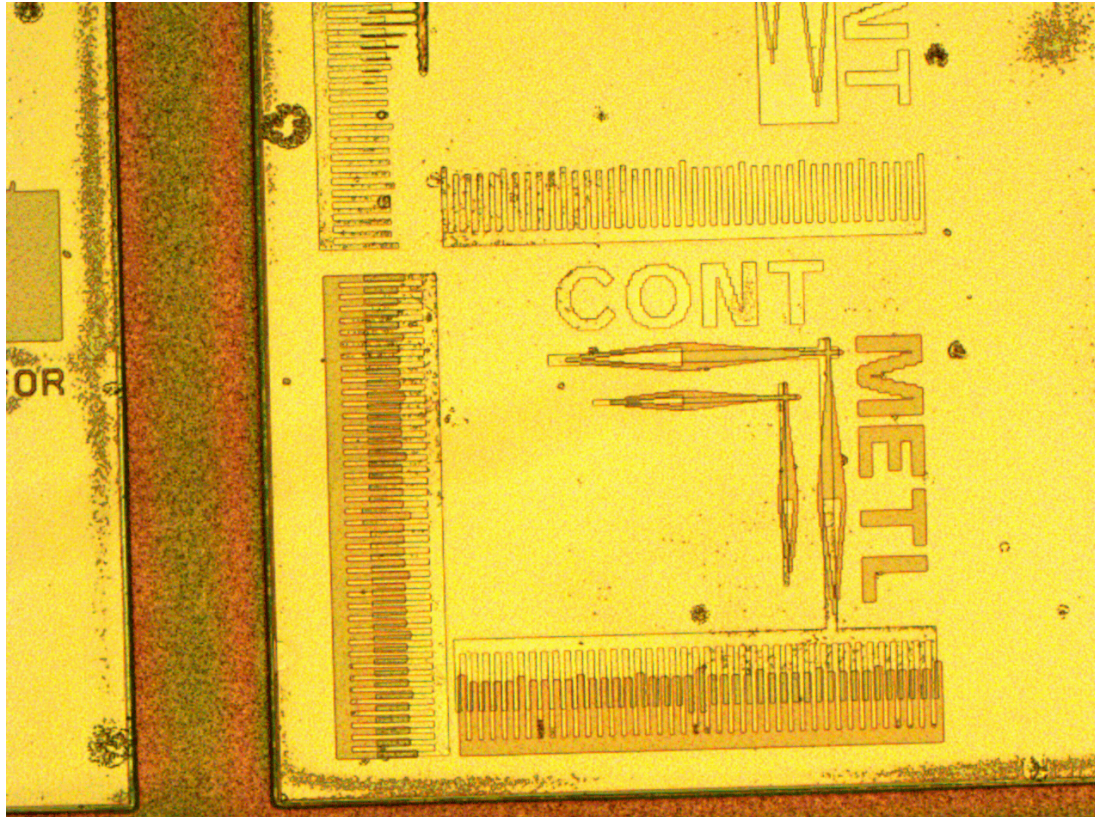


After lithography



After etch + strip

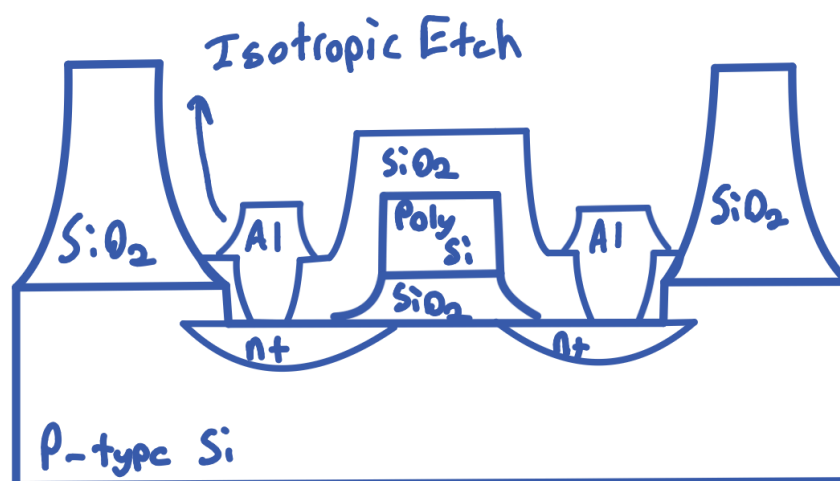
Overetch (μm)	2 μm	3 μm	4 μm	8 μm	average
After lithography	-0.132 μm	0.19 μm	0.338 μm	0.42 μm	0.204 μm
After etch + strip	-0.304 μm	-0.215 μm	-0.231 μm	-0.455 μm	-0.301 μm



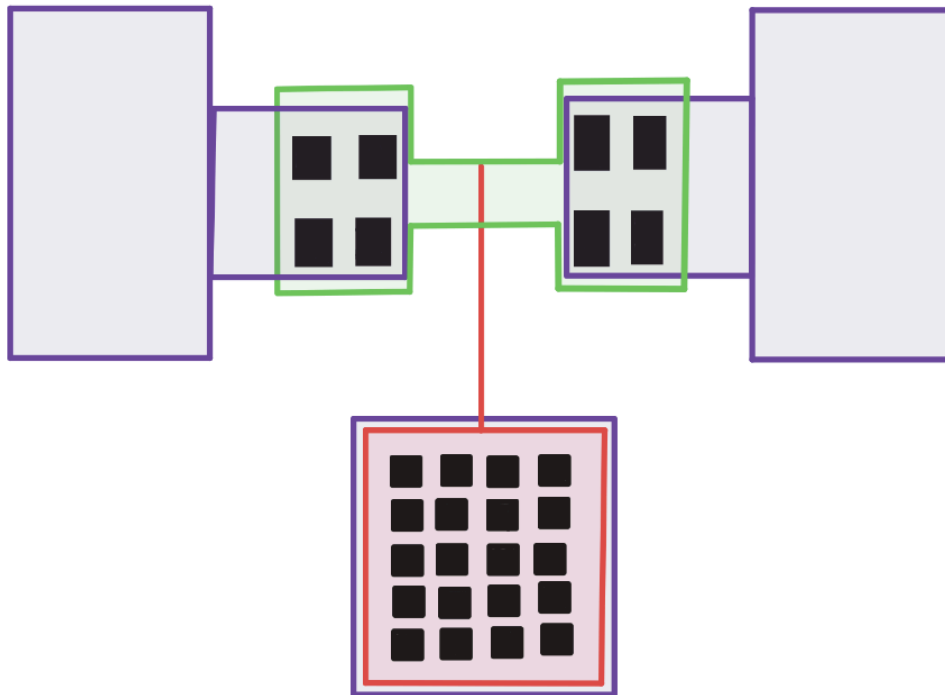
CONT/METL alignment mark

X misalignment (μm)	Y misalignment (μm)
0.6 μm	0.2 μm

METL Photolithography + Etch MOSFET Cross-Section Drawing:



METL Photolithography + Etch MOSFET Top-Down Drawing:



■ ACTV ■ POLY ■ CONT ■ METL

III. Theoretical Calculations

Oxide Thicknesses

Graphical Approach

Water <100> Silicon

Field oxide: dry - wet - dry $1000^{\circ}\text{C} \rightarrow$ wet should dominate
 $\rightarrow$ b/c 125 mins

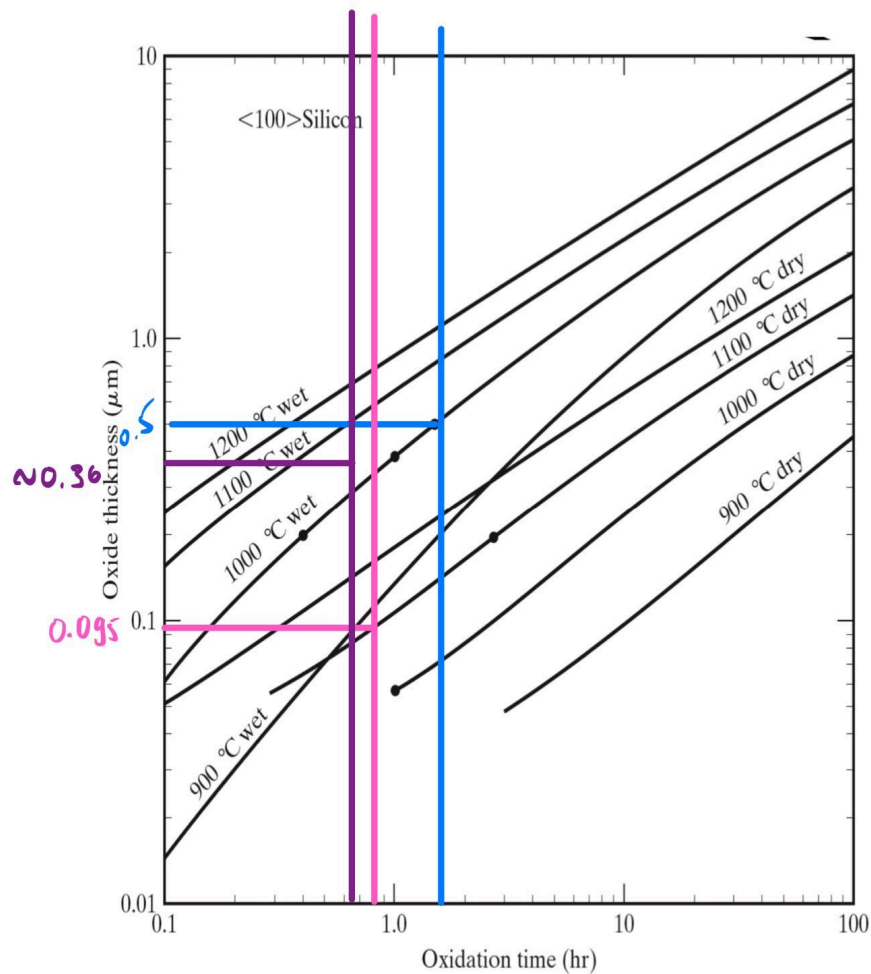
Gate oxide: dry oxidation at 1100°C

Intermediate oxide: wet oxidation at 1050°C

Field oxide $\approx 0.5 \mu\text{m} = 500 \text{ nm}$

Gate oxide $\approx 0.36 \mu\text{m} = 360 \text{ nm} \rightarrow$ Assumed 1050°C

Intermediate oxide $\approx 0.095 \mu\text{m} = 95 \text{ nm}$



Mathematical Approach:

We followed [Jaeger's method](#) (right side) as well as researched another approach for potentially more accurate results. In addition to this, we decided to also use "[Ch. Hollauer: Modeling of Thermal Oxidation and Stress Effects](#)" (left side). According to the resource, it is applicable to convert <111> silicon to <100> silicon just by multiplying it by $\frac{1}{1.68}$ as shown below, and use the given graph to find B and B/A values to use for our calculations. The computed results were very similar to the results obtained through the graphical approach.

$$\left(\frac{B}{A}\right)_{\langle 100 \rangle} = \frac{1}{1.68} \left(\frac{B}{A}\right)_{\langle 111 \rangle}$$

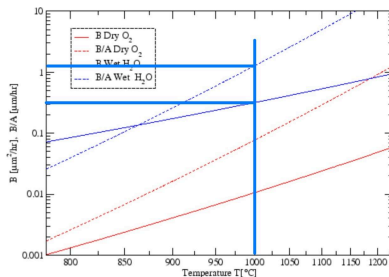
Field oxidation at 1000°C wet, 100 mins $t = 100 \text{ min} = 1.667 \text{ hrs}$

$$B = 0.31 \mu\text{m}^2/\text{hr} \quad \left(\frac{B}{A}\right)_{\langle 111 \rangle} \approx 1.25 \mu\text{m}/\text{hr}$$

$$\left(\frac{B}{A}\right)_{\langle 100 \rangle} = \frac{1.25}{1.68} = 0.744 \mu\text{m}/\text{hr} \quad A = \frac{B}{B/A} = \frac{0.31}{0.744} = 0.42 \mu\text{m}$$

$$Bt = (0.31)(1.667) = 0.523 \mu\text{m}^2$$

$$x_{ox} = \frac{-0.422 + \sqrt{(0.422)^2 + 4(0.523)}}{2} = 0.542 \mu\text{m} = 542 \text{ nm}$$



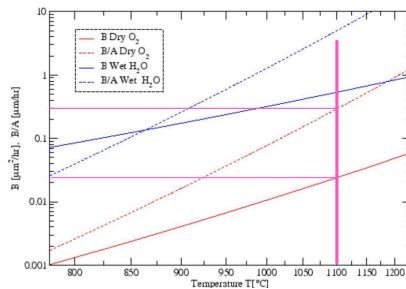
Gate oxidation at 1100°C dry, 50 mins $t = 50 \text{ min} = 0.833 \text{ hr}$

$$B \approx 0.025 \mu\text{m}^2/\text{hr} \quad \left(\frac{B}{A}\right)_{\langle 111 \rangle} \approx 0.3 \mu\text{m}/\text{hr}$$

$$\left(\frac{B}{A}\right)_{\langle 100 \rangle} = \frac{0.3}{1.68} = 0.1786 \mu\text{m}/\text{hr} \quad A = \frac{B}{B/A} = \frac{0.025}{0.1786} = 0.14 \mu\text{m}$$

$$Bt = (0.025)(0.833) = 0.0208 \mu\text{m}^2$$

$$x_{ox} = \frac{-0.14 + \sqrt{(0.14)^2 + 4(0.0208)}}{2} = 0.0903 \mu\text{m} = 90.3 \text{ nm}$$



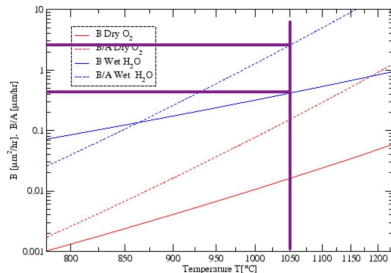
Intermediate oxidation at 1050°C wet, 40 mins

$$B \approx 0.43 \mu\text{m}^2/\text{hr} \quad \left(\frac{B}{A}\right)_{\langle 111 \rangle} \approx 2.5 \mu\text{m}/\text{hr}$$

$$\left(\frac{B}{A}\right)_{\langle 100 \rangle} = \frac{2.5}{1.68} = 1.49 \mu\text{m}/\text{hr} \quad A = \frac{B}{B/A} = \frac{0.43}{1.49} = 0.289 \mu\text{m}$$

$$Bt = (0.43)(0.667) = 0.287 \mu\text{m}^2$$

$$x_{ox} = \frac{-0.289 + \sqrt{(0.289)^2 + 4(0.287)}}{2} = 0.41 \mu\text{m} = 410 \text{ nm}$$



<100> silicon Field oxidation at 1000°C wet, 100 mins

$$T = 1000 + 273 = 1273 \text{ K} \quad t = 100/60 = 1.667 \text{ hr}$$

$$B = 386 \exp\left(-\frac{0.78}{KT}\right) = 386 \exp\left(-\frac{0.78}{(8.617 \times 10^{-5})(1273)}\right)$$

$$B = 0.315 \mu\text{m}^2/\text{hr}$$

$$\frac{B}{A} = 9.7 \times 10^7 \exp\left(-\frac{2.05}{KT}\right) = 9.7 \times 10^7 \exp\left(-\frac{2.05}{(8.617 \times 10^{-5})(1273)}\right)$$

$$\frac{B}{A} \approx 0.742 \mu\text{m}/\text{hr}$$

$$A = \frac{B}{B/A} = \frac{0.315}{0.742} = 0.425 \mu\text{m}$$

$$x_{ox} = \frac{-0.425 + \sqrt{(0.425)^2 + 4(0.315)(1.667)}}{2}$$

$$x_{ox} \approx 0.543 \mu\text{m} \approx 543 \text{ nm}$$

<100> silicon Gate oxidation at 1100°C dry, 50 mins

$$T = 1100 + 273 = 1373 \text{ K} \quad t = 50/60 = 0.833 \text{ hr}$$

$$B = 772 \exp\left(-\frac{1.23}{KT}\right) = 772 \exp\left(-\frac{1.23}{(8.617 \times 10^{-5})(1373)}\right)$$

$$B = 0.0236 \mu\text{m}^2/\text{hr}$$

$$\frac{B}{A} = 3.71 \times 10^6 \exp\left(-\frac{2.00}{KT}\right) = 3.71 \times 10^6 \exp\left(-\frac{2.00}{(8.617 \times 10^{-5})(1373)}\right)$$

$$\frac{B}{A} \approx 0.169 \mu\text{m}/\text{hr}$$

$$A = \frac{B}{B/A} = \frac{0.0236}{0.169} = 0.14 \mu\text{m}$$

$$x_{ox} = \frac{-0.14 + \sqrt{(0.14)^2 + 4(0.0236)(0.833)}}{2}$$

$$x_{ox} \approx 0.0868 \mu\text{m} \approx 86.8 \text{ nm}$$

<100> silicon Intermediate oxidation at 1050°C wet, 40 mins

$$T = 1050 + 273 = 1323 \text{ K} \quad t = 40/60 = 0.667 \text{ hr}$$

$$B = 386 \exp\left(-\frac{0.78}{KT}\right) = 386 \exp\left(-\frac{0.78}{(8.617 \times 10^{-5})(1323)}\right)$$

$$B = 0.412 \mu\text{m}^2/\text{hr}$$

$$\frac{B}{A} = 9.7 \times 10^7 \exp\left(-\frac{2.05}{KT}\right) = 9.7 \times 10^7 \exp\left(-\frac{2.05}{(8.617 \times 10^{-5})(1323)}\right)$$

$$\frac{B}{A} \approx 1.504 \mu\text{m}/\text{hr}$$

$$A = \frac{B}{B/A} = \frac{0.412}{1.504} = 0.274 \mu\text{m}$$

$$x_{ox} = \frac{-0.274 + \sqrt{(0.274)^2 + 4(0.412)(0.667)}}{2}$$

$$x_{ox} \approx 0.405 \mu\text{m} \approx 405 \text{ nm}$$

Parameter	Value [Jaeger's approach]
Field oxide (nm)	543 nm
Gate oxide (nm)	86.8 nm
Intermediate oxide (nm)	405 nm

Dopants

1) Peak Boron Concentration

Starting wafer (from process manual) - p-type,
initial resistivity = $14-16 \Omega \cdot \text{cm}$, $\langle 100 \rangle$ orientation,
blanket boron implantation of $6 \cdot 10^{12} \text{ cm}^{-2}$

Background Boron Concentration

$$\rho = \frac{1}{q_{\text{up}} N_A} \Rightarrow N_B = \frac{1}{q_{\text{up}} \rho} \Rightarrow$$

$$N_B = \frac{1}{1.602 \cdot 10^{-19} \cdot 500 \text{ cm}^2 \text{ V} \cdot 15.2 \text{ cm}} = 8.322 \cdot 10^{14}$$

Boron Implant:

$$N(x) = N_p \exp\left[-\left(\frac{x - R_p}{\Delta R_p}\right)^2 / 2\right]$$

$$N_p = \frac{Q}{\sqrt{2\pi} \Delta R_p}$$

$N(x)$ peaks at $x = R_p$, so

$$N_B = N_{B, \text{back}} + N_p$$

$$\Delta R_p \text{ for Boron at } 60 \text{ keV} \approx 0.05 \mu\text{m}$$

$$N_p = \frac{6 \cdot 10^{12}}{5.106 \text{ cm} \cdot \sqrt{2\pi}} = 4.787 \cdot 10^{17}$$

$$N_B = 4.787 \cdot 10^{17} + 8.332 \cdot 10^{14}$$

$$\approx 4.795 \cdot 10^{17} \text{ cm}^{-3}$$

Depth of peak Boron concentration at
 $x = R_p$.

$$R_p \text{ for B at } 60 \text{ keV} = 0.2 \mu\text{m}$$

Phosphorus:

$$T = 1050^\circ\text{C} \quad + \text{predep} = 10 \text{ min} \quad + \text{drive} = 37 \text{ min}$$

$$D_p = D \exp\left(-\frac{E_A}{kT}\right) = 10.5 \exp\left(-\frac{3.69}{kT}\right)$$

$$= 10.5 \exp\left(\frac{-3.69 \text{ eV}}{8.61733 \cdot 10^{-5} \text{ eV/K} \cdot (1050 + 273.15)}\right)$$

$$= 9.25 \cdot 10^{-14} \quad - \text{Model as constant source diffusion}$$

$$N(x, t) = N_0 \text{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Peak will be at surface concentration N_0 :

$$N_0 = \frac{Q}{\sqrt{\pi Dt}} \text{ but we do not have dose, use}$$

Solid Solubility to estimate.

$$N_0(P, 1050^\circ\text{C}) = 3.7 \cdot 10^{20} \text{ cm}^{-2}$$

$$Q = 2 N_0 \sqrt{\frac{D_p(t_{\text{depos}}) \cdot t_{\text{predep}}}{\pi}} \quad \begin{matrix} t_{\text{predep}} = 10 \text{ min} \\ = 600 \text{ s} \end{matrix}$$

$$= 2 \cdot 3.7 \cdot 10^{20} \cdot \sqrt{\frac{4.25 \cdot 10^{-14} \cdot 600}{\pi}}$$

$$= 3.11 \cdot 10^{15} \text{ cm}^{-2}$$

$$N(x, t) = \frac{Q}{\sqrt{\pi D t}} \exp\left(-\frac{x^2}{4 D t}\right) \quad \text{Peak at } x=0$$

$$\text{So } N_{p, \text{peak}} = \frac{Q}{\sqrt{\pi D t_{\text{drive}}}} \quad t_{\text{drive}} = 37 \text{ min} = 2220 \text{ s}$$

$$N_{p, \text{peak}} = \frac{3.11 \cdot 10^{15}}{\sqrt{\pi \cdot 4.25 \cdot 10^{-14} \cdot 2220}} = 1.224 \cdot 10^{20} \text{ cm}^{-2}$$

Peak phosphorus concentration at $x=0$

$$0 \text{ um}$$

$$x_j = 2 \sqrt{D t} + \sqrt{\ln\left(\frac{N_{p, \text{peak}}}{N_B}\right)}$$

$$x_j = 2 \sqrt{9.25 \cdot 10^{-14} \cdot 2220} + \sqrt{\ln\left(\frac{1.224 \cdot 10^{20}}{8.332 \cdot 10^{14}}\right)}$$

$$= 9.89 \cdot 10^{-5} \text{ cm}$$

$$= 0.989 \text{ um}$$

Lateral diffusion length:

$$L \sim \sqrt{D t}$$

$$\sqrt{D t_{\text{drive}}} = \sqrt{9.25 \cdot 10^{-14} \cdot 2220} = 1.43 \cdot 10^{-5} \text{ cm}$$

$$= 0.143 \text{ um}$$

Sheet resistance no phosphorus doping

$$R_s = \frac{\rho}{t} \quad \rho \approx [q \mu_p (N_A - N_D)]^{-1} \quad \text{p-type}$$

$$R_{s, \text{no P}} = \frac{\rho}{t} = \frac{15 \Omega \cdot \text{cm}}{0.38 \text{ mm}} = \frac{15 \Omega \text{ cm}}{0.038 \text{ cm}}$$

$$= 394.74 \Omega / \text{square}$$

Sheet resistance with P doping:

$$t \approx x_j = 0.989 \text{ um} = 9.89 \cdot 10^{-5} \text{ cm}$$

$$N \approx \frac{Q}{x_j} = \frac{3.11 \cdot 10^{15}}{9.89 \cdot 10^{-5}} = 3.145 \cdot 10^{19}$$

$$\rho = [1.602 \cdot 10^{-19} \cdot 500 \cdot 3.145 \cdot 10^{19}]^{-1} = 3.97 \cdot 10^{-4} \Omega \text{ cm}$$

$$R_{s, P} = \frac{3.97 \cdot 10^{-4}}{9.89 \cdot 10^{-5}} = 4.014 \Omega / \text{square}$$

Parameter	Value
Boron peak concentration (atoms/cm ³)	$4.795 * 10^{17} \text{ cm}^3$
Depth of boron peak concentration (μm)	0.2 μm
Phosphorus peak concentration (atoms/cm ³)	$1.224 * 10^{20} \text{ cm}^3$
Depth of phosphorus peak concentration (μm)	0 μm
Junction depth (μm)	0.989 μm
Sheet resistance of areas with no phosphorus doping (Ω/square)	394.75 Ω/square
Sheet resistance of areas with phosphorus doping (Ω/square)	4.014 Ω/square
Lateral diffusion of phosphorus (μm)	0.143 μm

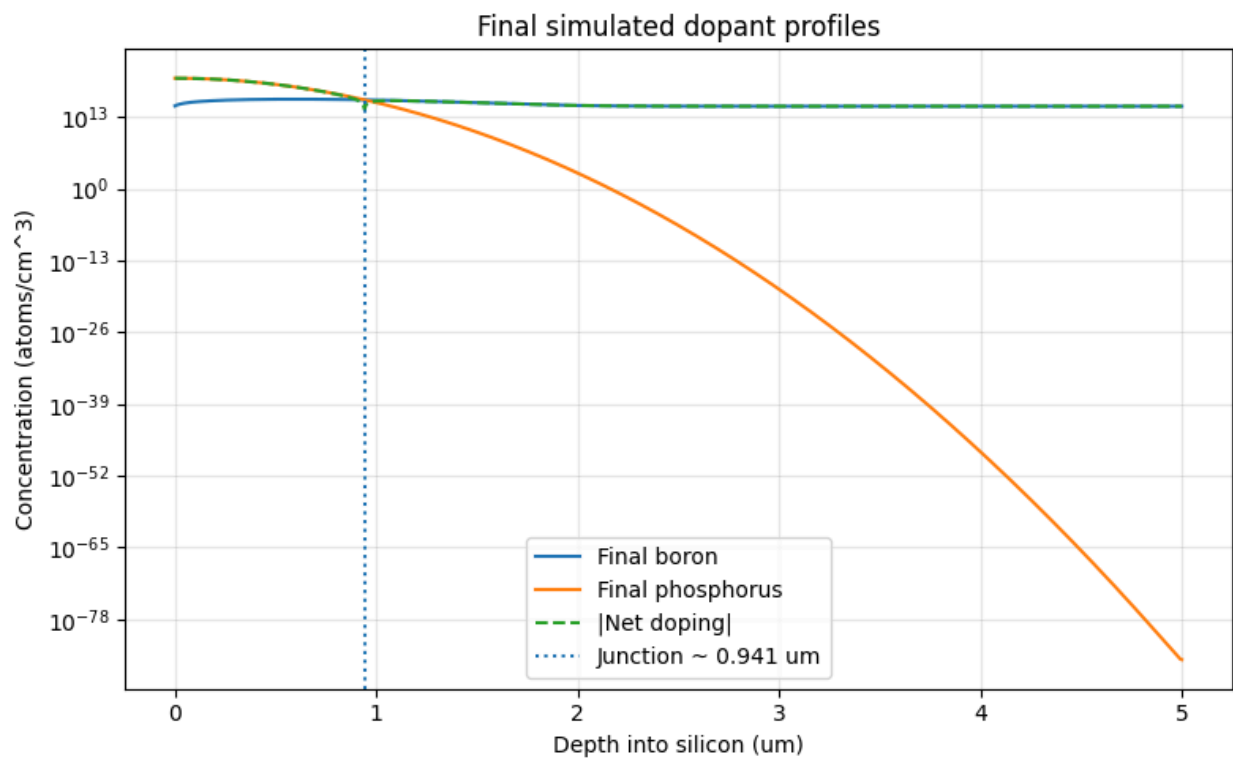
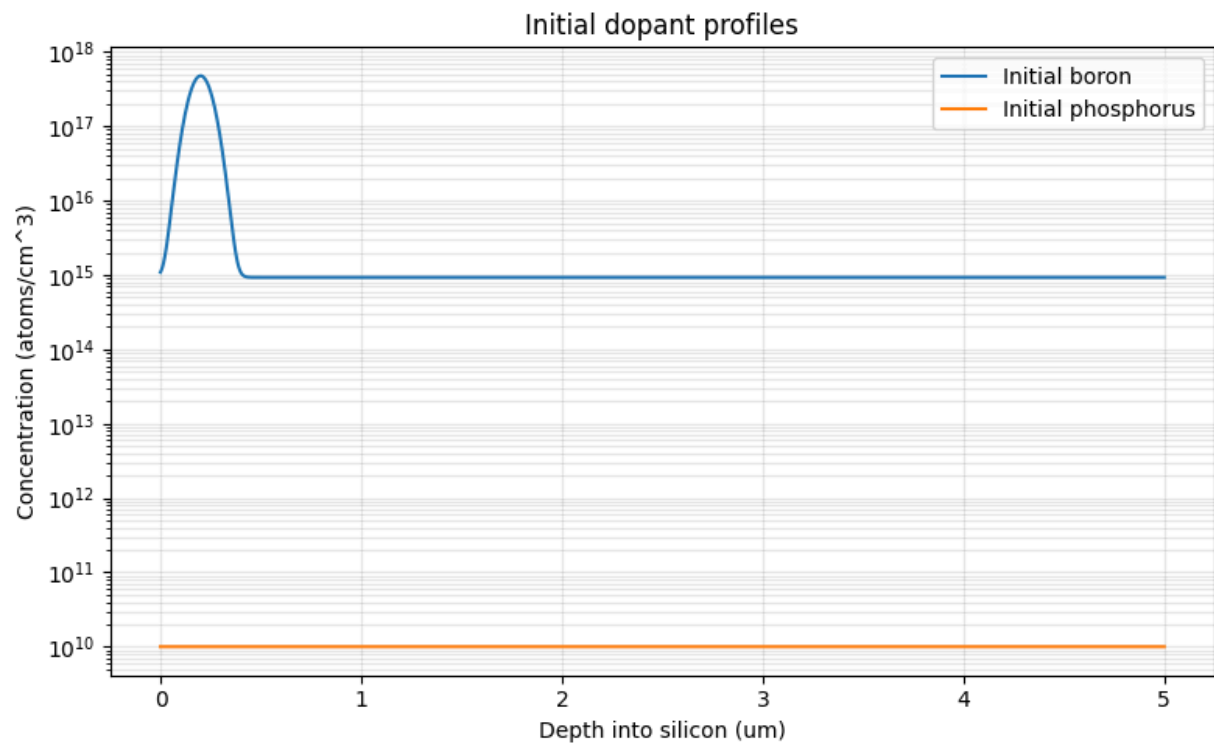
We also created a 1-D diffusion simulation to model the boron and phosphorus concentrations. The boron implant was initialized as a Gaussian profile, phosphorus predeposition was modeled using a fixed surface concentration, and we modeled the drive-in by solving the equations from the lecture numerically. To better reflect oxidation-induced boron depletion near the Si/SiO₂ interface, a simple surface-loss boundary condition was used during oxidation steps.

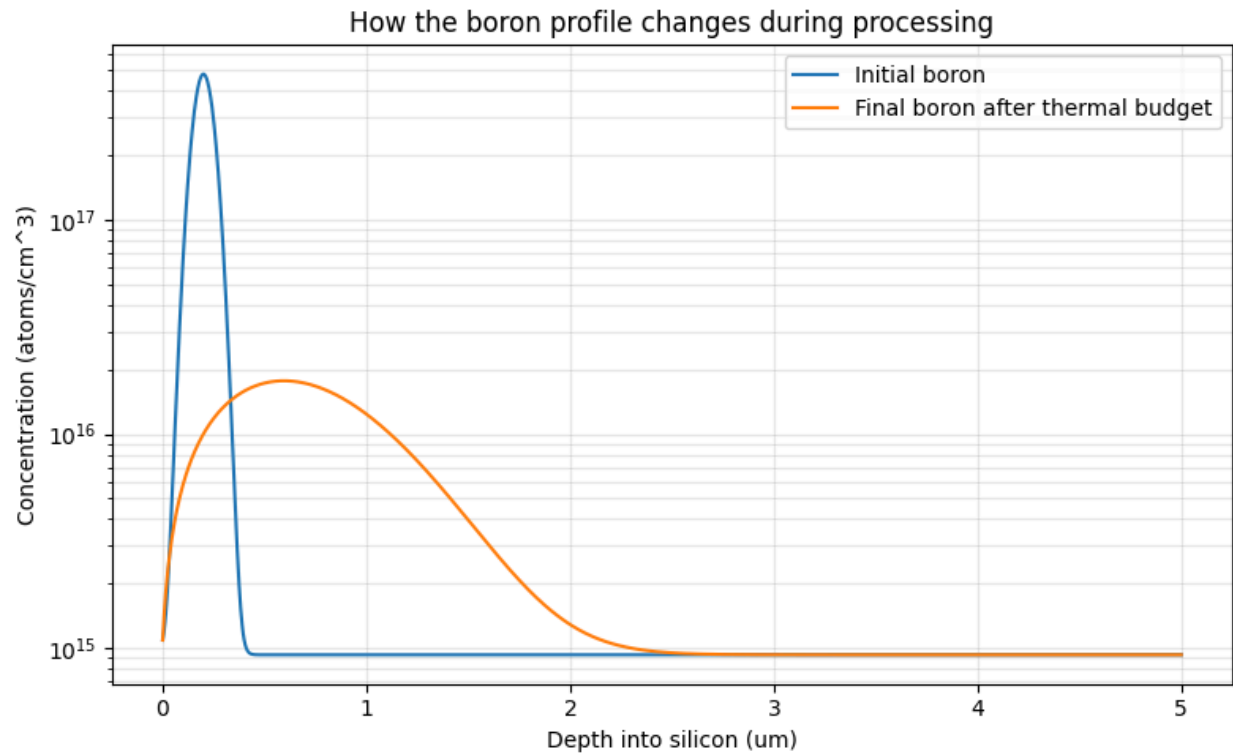
The results of the simulation are below:

```

==== EE143 dopant profile simulation ====
Boron peak concentration = 1.772e+16 cm^-3
Depth of boron peak = 0.5950 um
Phosphorus peak concentration = 1.129e+20 cm^-3
Depth of phosphorus peak = 0.0000 um
Junction depth = 0.9415 um
Lateral diffusion estimate = 0.2866 um

```





A sample of the code can be found [here](#).

Aluminum

$$R = \frac{\rho}{t} = (2.83 * 10^{-8} \Omega * m) / (100 * 10^{-9} m) = 0.283 \Omega/square$$

Aluminum sheet resistance (Ω/square)	0.283 Ω/square
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IV. Process Deviations (15 points)

Feature Size Deviation

Lithography step	Average overetch from lithography (μm)	Average overetch from wet etching (μm)
ACTV	$-0.0966 \mu\text{m}$	$0.244 - (-0.0966) = 0.3406 \mu\text{m}$
POLY	$0.189 \mu\text{m}$	$-0.456 - 0.189 = -0.645 \mu\text{m}$
CONT	$-0.2815 \mu\text{m}$	$-0.249 - (-0.2815) = 0.0325 \mu\text{m}$
METL	$0.204 \mu\text{m}$	$-0.301 - 0.204 = -0.505 \mu\text{m}$

The obtained results were not exact, partially impacted by factors such as contamination and overetching, affecting the overall impact of the lithography and etching. Due to the general dependency on timing, especially with regard to etching, we were extremely cautious during the process with the intent to avoid overetching. This resulted in slightly lower values than expected. Additionally, the non-idealities of the cleanroom and materials also raise the issue of contamination, as our wafers were subjected to issues like the door being left open and the usage of expired products.

Misalignment

Parameter	X misalignment (μm)	Y misalignment (μm)
ACTV-POLY	$0.6 \mu\text{m}$	$0.4 \mu\text{m}$
ACTV-CONT	$0.2 \mu\text{m}$	$0.2 \mu\text{m}$
CONT-POLY	$0.2 \mu\text{m}$	$0.2 \mu\text{m}$
CONT-METL	$0.6 \mu\text{m}$	$0.2 \mu\text{m}$

The misalignment in our results is not very significant, mainly affecting the ACTV-POLY and CONT-METL. Great care was taken during the mask alignment process to guard against misalignment in general. Despite this, there is still some mask misalignment, which is to be expected, as imperfections can result from human error and machine calibration. It is very possible that a small misalignment can go unnoticed, and can be exacerbated by the usage of sensitive older equipment, like that present in the cleanroom.

Oxide Growth

Oxidation step	Theoretical thickness (nm)	Measured thickness (nm)
Field oxidation	~543 nm	630 nm
Gate oxidation	~86.8 nm	110 nm
Intermediate oxidation	~405 nm	$(60.9 + 55)/2 = 57.96$ nm

The values were relatively close to each other, and the differences can be rationalized by taking into account the discrepancies in the materials that were limited for the sessions. In addition to this, non-idealities in baking can also affect the final product, as the wafer might be left to bake for longer than necessary. When conducting field and gate oxidation, the quantity of the photoresist was not a concern, but this progressively became more of an issue once we reached the intermediate oxidation step. Eventually, the photoresist needed to be rationed and split between the four wafers in our section, explaining the distinctive difference in the margin of error for intermediate oxidation in comparison to field and gate oxidation.

Doping

Doping profile	Theoretical value (Ω/square)	Measured value (Ω/square)
Sheet resistance before phosphorus doping	394.75 Ω/square	336 Ω/square
Sheet resistance after phosphorus doping	4.014 Ω/square	8.61 Ω/square

Doping is dependent on diffusion and thermal changes for a successful implementation. The doping process consisted of the usage of chemicals for a set amount of time, and was dependent on human ability to move wafers between the chemicals and rinsing steps, leaving room for potential errors during the process. Additionally, it is not necessarily guaranteed that the temperature was stable during the furnace and baking steps.

Aluminum

Aluminum	Theoretical value (Ω /square)	Measured value (Ω /square)
Sheet resistance	0.265	1.973

The aluminum varies based on the actual atmospheric pressure change in a vacuum, as this impacts the impurities and thickness of the aluminum. In addition, the turbopump is required to do more work on the wafer than the rough pump, which is not guaranteed during the process. Additionally, our section's TA mentioned that the machine was also somewhat broken in that the buttons to control the pumps did not work properly. This was assuaged by the fact that some internal controls and switches could be used to control the evaporator instead, but this is still a potential source of error due to the deviation from the process manual and the potential for the machine to be worn down by use. It is therefore possible that a combination of human error and the age of the evaporator, in addition to previous deviations, could have caused our measured value to be skewed from the theoretical value, and it could be inferred that this interfered with the scattering process of the aluminum.

V. Questions

1. We do lithography steps under yellow light only. What would happen if we exposed the wafer to white light or sunlight before development? What about after development? Would red light damage your process?
 - The photoresist is highly reactive to light exposure in UV and white light. If the wafer is prematurely exposed to either white light or sunlight it would affect the lithography process as it would cause inconsistencies in the photoresist pattern and cause development that are not part of the controlled process. Additionally, the results would be difficult to reproduce because the way the white light/sunlight is exposed could have variances based on the angles. However, if exposed to white light or sunlight post-development, it would have a negligible impact. If red light was utilized, then there would be no effect on the photoresist, similarly to yellow light. This is due to the wavelengths in yellow and red light being low energy whilst UV is high energy and makes such a vast impact to photoresist.
2. Describe our photoresist. Do we use positive or negative photoresist? I-line or G-line? What do I-line and G-line mean? Briefly describe how the photoresist responds to spinning, UV light exposure, and development.
 - The photoresist we utilize is a dark red solution (701 MiR) that has the exposed portions dissolve, and it is a positive photoresist because it becomes more soluble to light exposure and requires soft baking to remove solvent. Photoresist responds to spinning by moving outwardly through centrifugal force. G-line and I-line reference to UV-wavelengths in photolithography. While the G-line is typically 436nm and light-sensitive, the I-line is seen at the 365nm wavelength. When considering both, it is likely that our photoresist is I-line because it is capable of higher resolution.
3. What are the purposes of the 110°C pre-photoresist bake, HMDS, softbake, post-exposure bake, and hard bake?
 - The purpose of the 110°C pre-photoresist bake is to remove any moisture that is left on the wafer so that it has a completely dry surface in order to improve photoresist adhesion. HMDS can improve adhesion between silicon dioxide and photoresist because it forms a hydrophobic monolayer on SiO₂. Softbake helps evaporate solvent and improves the resistance quality for exposure. Post-exposure bake makes exposed regions easier to dissolve since it increases the solubility of the resistance in development due to an increase in chemical reaction. Hard bake's purpose is to turn resist into a more durable etch mask without deforming the patterns.
4. What is the difference between wet and dry oxidation, and why do we choose one for field oxidation and the other for gate oxidation?
 - Wet oxidation tends to be a faster process, but it can produce a lower quality of oxide, so it's more ideal to use it for thick field oxidation (~5000Å to 6000Å). On the other hand, dry oxidation is slower, but it will yield a higher quality of SiO₂

compared to wet oxidation, making it more ideal for gate oxidation since it is thinner ($\sim 700\text{\AA}$ to $900\text{\AA}$).

5. Why is it important to clean the wafer in HF immediately before depositing spin-on glass? Also, why do wafers need to be cleaned well before furnace steps?
 - Cleaning the wafer in HF immediately before depositing spin-on glass helps with the removal of native oxide so that dopants can properly diffuse directly into the silicon. Additionally, the wafers need to be cleaned before furnace steps because it prevents contaminants from diffusing at high temperature, and ensures uniform films. This is essential to prevent any foreign contamination in the diffusion process.
6. What is the difference between 5:1 HF and 10:1 HF, and why do we need both for different steps? Also, the HF we use in the lab is actually buffered HF (a mixture of pure HF and NH_4F); why is this?
 - The 5:1 HF etches faster for thick oxides but has less control which is why it is used in our process for thick oxide removal. Whereas 10:1 HF etches slower but has a much better control (commonly used for thin oxide cases for stronger control). We use buffered HF because NH_4F actually helps with stabilizing HF better, giving us more control over the process as well as a more uniform oxide etch rate.
7. When designing a new microfabrication process, how do you know how long etch steps should be? Provide at least two different methods.
 - There are two forms in which etching takes place: dry and wet etching. When utilizing dry etching, it is important to utilize ovens with controllable high temperatures where gases can be processed onto the wafers evenly. Etch is calculated by the depth you want to obtain divided by the rate at which the chemical etches, to know the time it takes to etch. When utilizing wet etch, it is important to see that the penetrating acid used is taking visual effect on the wafer, which is visible through the back, resembling a rainbow before it turns white. Once the visual effect is seen at the center of the wafer after agitation, remove it.
8. Why does the aluminum evaporator have two vacuum pumps: a roughing pump and turbopump? Describe how they are used.
 - A roughing pump is used in order to achieve the vacuum pressure of 10^{-3} torr. A roughing pump displaces gas with the purpose of reaching the vacuum pressure. The turbopump is in charge of bringing the pressure down further so that the aluminum can evaporate at 10^{-7} torr. The mechanism utilized for this is spinning at high speed to push gas molecules toward the bottom of the mechanism, which should have the roughing pumps' effect and be kept down and away from the center. This is what will cause the evaporation to take place.
9. Please describe how our following metrology tools work and what they are useful for: the optical microscope, four-point probe, and Nanospec thin film measurement tool.
 - The optical microscope is imperative for using visible light and magnification lenses to check for defects and alignment. This is critical for determining if the wafer got masked and etched adequately. We would put the wafer under the microscope and be able to get images of our progress and how well the steps we

did that day worked. The four-point probe is useful for determining the sheet resistances that are being outputted from the wafer's conductive components. The four-point probe works by placing two outer probes with current and two inner with voltage to get the sheet resistance which presents the uniformity in the doping process and the diffusion accuracy. Then the Nanospec thin film measurement tool is useful for determining the thickness in layers produced throughout the microfabrication process. If the etching was done accurately enough the layers will be within the expected range for the specific steps and then it would be compared to the thickness you get from the Nanospec thin film measurement tool. This tool also works by referencing a control wafer to compare it to as it reflects the desirable refractive index and with the use of optical techniques such as constructive and destructive interference to calculate the thickness.

10. For two different processes, please explain how they could be replaced with a different process you learned about in lecture, and the pros and cons of doing so (e.g., replacing spin-on glass with ion implantation).
 - Thermal diffusion can be replaced with ion implantation and annealing in order to achieve the same results. Ion implantation creates more precision in the depth because implantation energy and dopant dosage are independently controlled factors and its accuracy is imperative for replicating the results. The con for ion implantation is that it forces you to anneal, it can cause damage in the lattice, and it can take longer to do the process. Another process that can be replaced is Physical Vapor Deposition (PVD) which includes sputtering and evaporation for the process of Chemical Vapor Deposition (CVD) oxide, which is advantageous for thick layers and its process is generally quicker and requires less hand-on approaches. Due to CVD utilizing chemicals primarily, it does require higher temperatures for the process and more advanced chemical reactions to the PVD methods, but because it creates chemical bonds it is generally viewed as stronger.

Resources

TABLE 3.1 Values for Coefficient D_0 and Activation Energy E_A for Wet and Dry Oxygen*

	Wet O ₂ (X _i = 0 nm)		Dry O ₂ (X _i = 25 nm)	
	D ₀	E _A	D ₀	E _A
<100> Silicon				
Linear (B/A)	$9.70 \times 10^7 \mu\text{m/hr}$	2.05 eV	$3.71 \times 10^6 \mu\text{m/hr}$	2.00 eV
Parabolic (B)	$386 \mu\text{m}^2/\text{hr}$	0.78 eV	$772 \mu\text{m}^2/\text{hr}$	1.23 eV
<111> Silicon				
Linear (B/A)	$1.63 \times 10^8 \mu\text{m/hr}$	2.05 eV	$6.23 \times 10^6 \mu\text{m/hr}$	2.00 eV
Parabolic (B)	$386 \mu\text{m}^2/\text{hr}$	0.78 eV	$772 \mu\text{m}^2/\text{hr}$	1.23 eV

*Data from Ref.[9]

EE143_Dopant_Profile_Sim, Google Colab.

<https://colab.research.google.com/drive/1C95WhcUo3YdpfSYkyu1iiqzl6QPQRoHn?usp=sharing>

Neudeck, G., & Pierret, R. (n.d.). *MODULAR SERIES ON SOLID STATE DEVICES*

Introduction to Microelectronic Fabrication.

<https://www.eng.auburn.edu/~jaeger/NewTOC.pdf>

Nondestructive Evaluation Techniques : Eddy Current Testing. (2017). Nde-Ed.Org.

https://www.nde-ed.org/NDETechniques/EddyCurrent/ET_Tables/ET_matlprop_Aluminum.xhtml

Introduction to G-Line and I-Line Photoresists. (2025). Youliantrade.Com.

<http://www.youliantrade.com/news/198.html>

arun2386. (2026). *Introduction to Microelectronic Fabrication R C Jaeger.* Scribd.

<https://www.scribd.com/doc/310146320/Introduction-to-Microelectronic-Fabrication-R-C-Jaeger>

2.6 *The Deal-Grove Model.* (2026). Tuwien.Ac.At.

<https://www.iue.tuwien.ac.at/phd/hollauer/node16.html#bbapara>